

Chapter 11

1) *p: true

&p: false

*&p: false

&*p: true

*i: false

&i: true

*&i: true

&*i: false

3)

```
void avg_sum(double a[], int n, double *avg, double *sum)
```

```
{
```

```
    int i;
```

```
    *sum = 0.0;
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        *sum += a[i];
```

```
    }
```

```
    *avg = *sum / n;
```

```
}
```

4)

```
void swap(int *p, int *q)
```

```
{
```

```
    int temp;
```

```
    temp = *p;
```

```
    *p = *q;
```

```
    *q = temp;
```

```
}
```

Chapter 12

1) $*(p + 3)$: 14

$*(q - 3)$: 34

$q - p$: 4

$p < q$: true

$*p < *q$: false

4)

```
#include <stdbool.h>
```

```
#define STACK_SIZE 100
```

```
int contents[STACK_SIZE];
```

```
int *top_ptr = contents;
```

```
void make_empty(void)
```

```
{
```

```
    top_ptr = contents;
```

```
}
```

```
bool is_empty(void)
```

```
{
```

```
    return top_ptr == contents;
```

```
}
```

```
bool is_full(void)
```

```
{
```

```
    return top_ptr == contents + STACK_SIZE;
```

```
}
```

```
void push(int i)
```

```
{
```

```
    if (is_full())
```

```
        stack_overflow();
```

```
    else
        *top_ptr++ = i;
}
```

```
int pop(void)
{
    if (is_empty())
        stack_underflow();
    else
        return *--top_ptr;
}
```

9)

```
double inner_product(const double *a, const double *b, int n)
{
    double sum = 0.0;

    for (int i = 0; i < n; i++) {
        sum += *(a + i) * *(b + i);
    }

    return sum;
}
```